Can Vision Models Read? Probing Linguistic Structure via JEPA on Text Screenshots

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Abstract

We ask whether vision models acquire linguistic structure when their training data consists of rendered text. As a probe, we use retrieval over a 2 000-page held-out slice of Wikipedia screenshots: given a 224×224 crop, identify its source page. We compare six encoders trained with five distinct objectives, including a joint-embedding predictive architecture (I-JEPA) that is the canonical image-only predictive baseline. The finding is sharp: image–text contrastive encoders (CLIP, SigLIP) retrieve at $\mathrm{R@10} = 0.78$ and 0.62 zero-shot, while image-only predictive (I-JEPA), self-supervised contrastive (DINOv2), supervised classification (ImageNet ViT), and random-init plain ViT all sit at $\mathrm{R@10} \leq 0.058$. I-JEPA in particular scores *below* random initialisation. Across the six encoders, mutual-kNN alignment with text encoders rank-correlates with retrieval R@10 at Spearman $\rho = 0.83$ ($p = 0.04$), supporting the Platonic Representation Hypothesis as the explanatory axis. Fine-tuning DINOv2 with a SALAD aggregator on 50 k pages reaches R@10 $= 0.915$, exceeding zero-shot CLIP, but every fine-tuned checkpoint exhibits a layout-fingerprint failure mode: erasing the top 15–30 % of every test image drops in-distribution Protocol-A R@10 by up to 0.29 while *improving* cross-set Phase-2 R@1 by up to $6.7\times$ on the same checkpoints. Zero-shot CLIP and SigLIP move in the opposite direction (R@10 *rises* by 0.07–0.11 under blanking), falsifying the title-OCR explanation for their zero-shot edge. A linear adapter trained on (image, BERT-of-title) pairs lifts I-JEPA from $\mathrm{R@10} = 0.015$ to 0.197, recovering latent text alignment that is invisible without the projection.

1. Introduction

Documents combine glyph sequences with visual layout. A vision encoder trained on rendered text could in principle acquire two distinct kinds of structure: (i) glyph-level reading (recognising words and characters as visual patterns), and (ii) layout-level structure (titles, infoboxes, lists, figures, captions). The two pipelines in Figure 1 operationalise the alternatives: the text pipeline OCRs the page and encodes characters; the visual pipeline encodes the screenshot directly. The visual pipeline is attractive when text extraction is unreliable (degraded scans, formula-heavy or multilingual content) or when the query is itself a visual crop with no text available, but it places the burden of "reading" on the encoder.

We probe whether modern vision encoders read in this sense by measuring how well they retrieve Wikipedia source pages from 224×224 crops. The probe is informative because the visual content *is* the text: an encoder that genuinely reads should match an encoder that operates on the underlying string. We compare five families:

- **Image–text contrastive** (CLIP, SigLIP) — learns by matching images and captions; sees text indirectly.
- **Image-only predictive** (I-JEPA) — the canonical joint-embedding predictive architecture; trains an image encoder to predict the latent representations of masked image blocks (**?**). No text supervision.
- **Image-only contrastive** (DINOv2) — view-contrastive distillation on natural images. No text supervision.
- **Supervised classification** (ImageNet ViT) — ViT-B/16 trained on ImageNet-21k+1k labels.
- **Random initialisation** — a frozen ViT-B/16 sanity floor.

Three questions:

Q1 (zero-shot reading). Which pretraining produces image representations from which the source page can be retrieved zero-shot?

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COMP547 Deep Unsupervised Learning, Spring 2025.

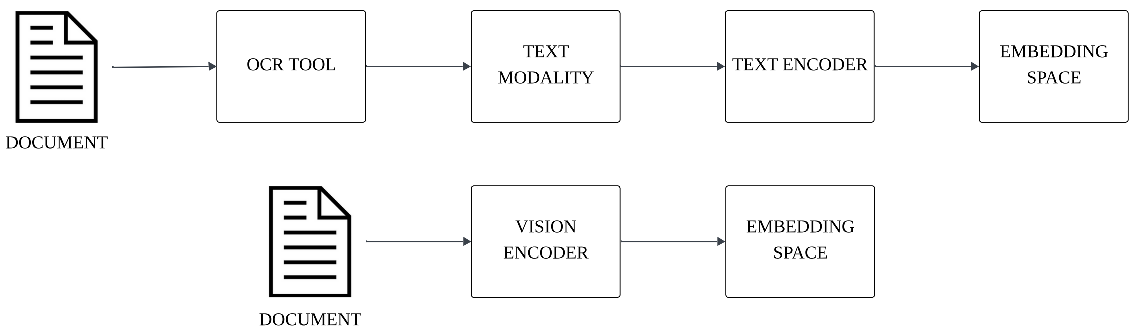

Figure 1. The two retrieval paradigms compared in this paper. The *text* pipeline (top) extracts characters with an OCR tool and encodes them with a text encoder. The *visual* pipeline (bottom) feeds the rendered document image directly into a vision encoder. We study the visual pipeline because it bypasses OCR errors, multilingual tokeniser failures, and layout discarding, and because it admits image queries (e.g. a crop of an unknown page) for which no text exists.

Q2 (JEPA specifically). Does I-JEPA, an image-only predictive objective with no text supervision, acquire any linguistic structure detectable through retrieval?

Q3 (the Platonic axis). Does the Platonic Representation Hypothesis (**?**) — which predicts that encoders trained on diverse objectives converge to a shared abstract representation — explain the retrieval ordering quantitatively? We measure each image encoder's mutual-kNN alignment with text encoders on the same content and ask whether alignment predicts retrieval performance.

Contributions. (1) A leak-free crop→page protocol with leave-one-out gallery aggregation. (2) A six-encoder zero-shot grid showing image–text contrastive encoders dominating the four image-only baselines, with I-JEPA sitting *below* random initialisation. (3) A Platonic alignment grid producing the rank correlation $\rho = 0.83$ between alignment and retrieval, statistically significant at $n = 6$. (4) A reproducible failure mode in every fine-tuned checkpoint: erasing the top 15–30 % of every test image *improves* cross-set transfer, indicating the model overweights a layout confound rather than reading content.

2. Related Work

Pixel-based language models and document AI. PIXEL (**?**), CLIPPO (**?**) and Pix2Struct (**?**) all train transformers on rendered text and demonstrate that pixel inputs can match or exceed tokenised text on downstream tasks. ColPali (**?**) produces multi-vector page embeddings via late interaction and shows that direct visual retrieval over screenshots competes with text+OCR pipelines on the ViDoRe benchmark. We adopt a simpler single-vector setup to keep our six encoders comparable on identical infrastructure.

Self-supervised vision and post-hoc text alignment. DINOv2 (**?**) learns image features via view-contrastive distillation; I-JEPA (**?**) learns them by predicting masked-block representations. Jose et al. (**?**) showed that a small text encoder aligned post-hoc to a frozen DINOv2 reaches CLIP-level zero-shot classification, suggesting that self-supervised image features already contain a linguistically-decodable structure. We test the analogous claim for I-JEPA on document screenshots.

Image–text contrastive pretraining and OCR shortcuts. CLIP (**?**) and SigLIP (**?**) are trained contrastively on image–caption pairs, with CLIP using softmax InfoNCE and SigLIP using a sigmoid pairwise loss. Both have been shown to read text rendered in images (**??**); we exploit this in a title-blanking experiment.

Platonic alignment and probing. Huh et al. (**?**) predict cross-encoder alignment grows with model and data scale; Maniparambil et al. (**?**) and Moayeri et al. (**?**) measure such alignment for natural-image vision encoders. We extend this measurement to a document-screenshot domain and use mutual-kNN, debiased CKA (**??**), and Procrustes distance on identical 2 000-page eval slices.

3. Method

3.1. Data and crops

We use `Tevatron/wiki-ss-corpus`: ~1.2 M Wikipedia page screenshots paired with the page title and rendered body text. We hold a local cache of ~376 k pages. A non-overlapping cropper tiles each page into 490×490 px crops resized to 224×224 px. At this resolution body text is sub-pixel and illegible to a ViT patch encoder; only headings and the overall layout structure remain visually recoverable. "Reading" in this paper therefore refers to heading-level text and visual layout, not fluent body content.

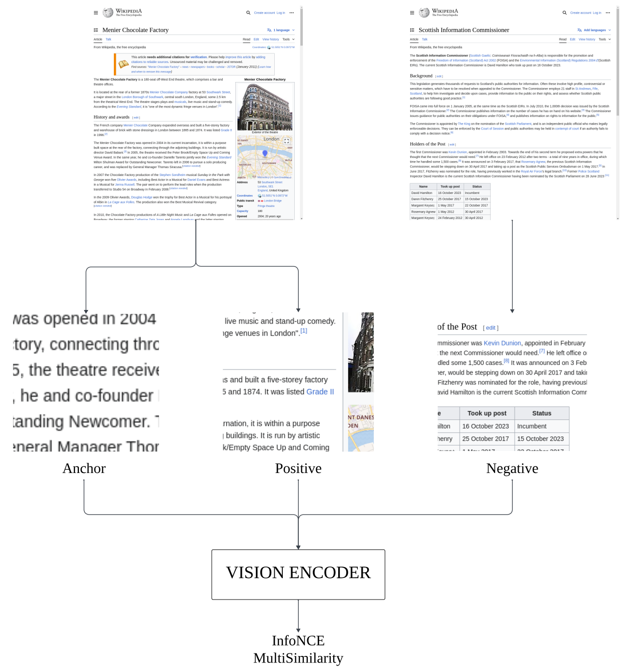

Figure 2. Training. Each step samples four pages × four crops; same-page crops are positives, cross-page crops are negatives. The image encoder produces an L2-normalised descriptor; training minimises a multi-similarity contrastive loss (**?**). We unfreeze the last four ViT blocks (of twelve) and train a head on top.

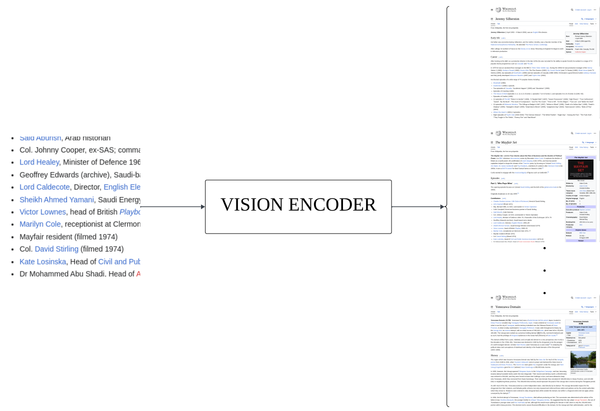

Figure 3. Anchor-pool benchmark (Phase 2). Each anchor crop is encoded with the vision encoder and ranked against a small pool ($\sim$7 images) of positives (other crops of the same page) and hard negatives (crops of unrelated pages).

3.2. Training (fine-tunes)

We train each (encoder, head) configuration with the procedure of Figure 2. Each batch contains 4 pages $\times$ 4 crops $= 16$ images. Within a batch, crops sharing a page are positives; otherwise negatives. The loss is multi-similarity (**?**). The optimiser is AdamW with weight decay 10^{-4}, gradient clip 1.0, linear warmup over 5 % of steps followed by cosine decay. Backbone learning rate is 10^{-5} for DINOv2 and 10^{-7} for CLIP and SigLIP (a higher rate causes catastrophic representation collapse on the latter two), head learning rate 5×10^{-4} throughout. Heads are either an MLP ($768 \to 512 \to 256$) or the SALAD optimal-transport aggregator (**?**) producing an 8 448-d descriptor. Only the head and the last four ViT blocks are trainable. Fine-tunes use 30 000 training pages $\times$ 3 epochs (5 625 optimiser steps), one seed; the saturation diagnostic uses 50 000 pages.

3.3. Evaluation

We evaluate on a held-out slice of $P = 2\,000$ pages disjoint from training, with two complementary protocols.

Protocol A — crop → page (headline metric). For each query crop c on page p with crops $\{c_1, \ldots, c_{n_p}\}$, the gallery vector for any page $q \neq p$ is the L2-normalised mean of all q's crop embeddings. The gallery vector for page p is recomputed leaving the query out: $g_p^{\setminus c} = \text{normalise}\big(\sum_i \phi(c_i) - \phi(c)\big)$. We cosine-rank the P gallery vectors against $\phi(c)$ and report R@k as the fraction of queries whose source page lands in the top k. Random R@1 $= 1/P = 5 \times 10^{-4}$. The leave-one-out aggregation forbids the trivial "find another crop of the same column" shortcut implicit in earlier crop-vs-crop formulations of this task.

Protocol B — anchor-pool transfer. The anchor-pool benchmark of Figure 3 uses 118 valid validation triplets from `Wikipedia_SS_withanchors`. For each anchor crop, the pool is the union of its positives and hard negatives ($\sim$7 images). We rank the pool by cosine similarity and report R@1. This measures cross-set transfer: anchors and pool images come from a different distribution of crops than what the model trained on.

3.4. Platonic alignment

For each (image encoder E_v, text encoder E_t) pair we encode the same 2 000-page eval slice with both halves: image features $\mathbf{X} \in R^{N \times d_v}$ from page screenshots, text features $\mathbf{Y} \in R^{N \times d_t}$ from page titles. We report mutual-kNN at $k = 10$ (the fraction of points sharing at least one of their top-10 neighbours under the two embeddings), debiased CKA (**??**), and Procrustes distance. The Platonic Hypothesis predicts a positive rank correlation between E_v's alignment and E_v's downstream cross-modal retrieval performance. We test this directly in §4.

Table 1. Protocol-A zero-shot retrieval at $P = 2000$. Random R@1 $= 5 \times 10^{-4}$. Encoder rows are coloured by pretraining family: image–text contrastive image-only self-supervised supervised random . "Gap" is same-page minus diff-page mean cosine similarity.

Encoder	R@1	R@5	R@10	R@20	gap
CLIP ViT-B/16	**0.548**	0.715	**0.779**	0.833	$+0.205$
SigLIP ViT-B/16	0.348	0.541	0.624	0.703	$+0.110$
DINOv2 ViT-B/14	0.022	0.043	0.058	0.075	$+0.010$
I-JEPA ViT-H/14	0.002	0.008	0.015	0.021	-0.002
ImageNet ViT-B/16	0.009	0.022	0.031	0.043	-0.002
Plain ViT-B/16	0.007	0.020	0.029	0.042	$+0.000$

Table 2. Protocol-A retrieval after last-4-block fine-tuning, single seed. "ΔZS" is R@10 minus the encoder's zero-shot baseline. Yellow rows are fine-tuned variants of the encoder above them.

Encoder + head	R@1	R@10	ΔZS
DINOv2 + MLP (30k)	0.351	0.787	$+0.729$
DINOv2 + SALAD (30k)	0.512	0.880	$+0.822$
DINOv2 + SALAD (50k)	**0.569**	**0.915**	$+0.857$
CLIP + SALAD (30k)	0.601	0.724	-0.055
CLIP + MLP (30k)†	0.000	0.000	collapsed

† Embeddings collapsed to a constant.

4. Results

4.1. Zero-shot retrieval

Table 1 reports zero-shot Protocol-A. CLIP dominates at R@10 $= 0.779$, SigLIP follows at 0.624, and the four image-only encoders cluster near random (R@10 $= 0.015$–0.058). Three observations follow:

(i) The largest gap is between the image–text contrastive and the image-only families (ΔR@10 ≈ 0.57), not between encoders within a family. The supervised ImageNet ViT scores 0.031 versus 0.029 for the random-init plain ViT, so ImageNet classification supervision adds essentially nothing for this task.

(ii) I-JEPA scores R@10 $= 0.015$, *below* the random-init plain ViT (0.029). Despite ViT-H/14 being roughly 5× larger than the ViT-B encoders, image-only predictive pretraining out-of-the-box does not transfer to visual document retrieval. Q2's optimistic reading is therefore falsified at the zero-shot level: a post-hoc text adapter (§5) is the only remaining path to rescue I-JEPA.

(iii) The CLIP–SigLIP gap (0.155) is real but small relative to the CLIP–DINOv2 gap (0.721). The dominant predictor is whether the encoder saw paired image–text data, not the specific contrastive form (softmax vs. sigmoid).

Table 3. Mutual-kNN@10 between image and text encoders on the same 2 000-page eval slice. Bold = highest entry per row. Row colours match Table 1.

image \ text	BERT	MiniLM	CLIP-T	SigLIP-T
CLIP	0.094	0.120	**0.138**	0.132
SigLIP	0.091	0.108	0.123	**0.131**
DINOv2	**0.027**	0.021	0.019	0.026
I-JEPA	**0.019**	0.014	0.016	0.018
ImageNet ViT	0.018	0.013	0.013	**0.019**
Plain ViT (rand)	0.008	0.007	0.007	**0.009**

4.2. Fine-tuning

Table 2 reports the fine-tuned grid. DINOv2 + SALAD on 30 k pages reaches R@10 $= 0.880$ and on 50 k pages reaches 0.915, exceeding zero-shot CLIP (0.779). The MLP head on the same DINOv2 backbone scores 0.787, so the SALAD aggregator contributes $+9.3$ pp R@10 over an MLP at the matched backbone — consistent with the prior place-recognition result of (**?**).

CLIP fine-tuning at the conservative $\eta_{\text{bb}} = 10^{-7}$ hurts more than it helps. CLIP+SALAD loses 5.5 pp R@10 from its own zero-shot baseline; the same-page minus diff-page sanity gap shrinks from $+0.205$ to $+0.032$, indicating the fine-tune sharpened the top-1 match (R@1 $+5.3$ pp) but compressed the overall similarity geometry. CLIP+MLP collapsed to a constant output by step 400. Training CLIP at the LR that worked for DINOv2 (10^{-5}) caused immediate divergence in earlier attempts; 10^{-7} was chosen to avoid that, but evidently the CLIP+MLP combination cannot navigate the resulting flat-loss basin. A multi-LR sweep is the proper next step but is deferred.

4.3. Platonic alignment predicts retrieval

Table 3 reports image-encoder × text-encoder mutual-kNN@10 alignment. CLIP and SigLIP align with text encoders at 0.09–0.14; the four image-only encoders align at 0.007–0.027. Figure 4 plots, for each image encoder, its column-maximum text alignment against its zero-shot Protocol-A R@10. The Spearman correlation is 0.83 ($p = 0.04$, $n = 6$): statistically significant even at this small sample size. The two image–text contrastive encoders sit in the upper-right corner; the four image-only encoders sit in the lower-left.

This is the central empirical claim of the paper. The kind of pretraining the encoder received is a strong predictor of its retrieval performance, mediated by its alignment with text representations on the same content. The Platonic Hypothesis is supported in this domain: encoders close to the text-encoder geometry retrieve text-heavy visual documents better.

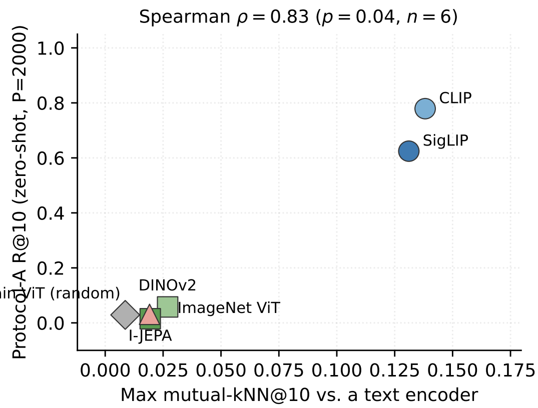

Figure 4. Each point is one image encoder. The x-axis is its maximum mutual-kNN@10 against the four text encoders (column max of Table 3); the y-axis is its zero-shot Protocol-A R@10 (Table 1). Marker shape and pastel colour code the pretraining family. Spearman $\rho = 0.83$ ($p = 0.04$, $n = 6$).

Table 4. Protocol-A R@10 with the top 15 % of every page painted white before cropping (b-15) versus the original eval (orig.). Top block: zero-shot encoders. Bottom block: fine-tuned checkpoints. Sign of Δ is the central observation: zero-shot CLIP/SigLIP *gain* from blanking (negative confound removed); trained DINOv2 *loses* substantially (the model learned to depend on the title region).

Encoder	orig. R@10	b-15 R@10	Δ
CLIP zero-shot	0.779	0.852	+0.073
SigLIP zero-shot	0.624	0.731	+0.106
DINOv2 zero-shot	0.058	0.058	−0.000
I-JEPA zero-shot	0.015	0.012	−0.003
ImageNet zero-shot	0.031	0.032	+0.001
Plain-ViT zero-shot	0.029	0.030	+0.001
DINOv2-MLP-30k	0.787	0.494	−0.292
DINOv2-SALAD-30k	0.880	0.720	−0.161
DINOv2-SALAD-50k	0.915	0.673	−0.242
CLIP-SALAD-30k	0.724	0.687	−0.037

4.4. Title-region ablation

We test which pixels each model uses by re-encoding the same 2 000-page eval slice with the top 15 % (and 30 %) of every page painted white before cropping, then re-running both protocols. Tables 4 and 5 report the deltas, and the three rows of the result tell different stories.

Zero-shot image–text encoders gain from blanking. CLIP and SigLIP both improve under title removal (R@10 +0.073 and +0.106 respectively); the four image-only encoders move by ≤ 0.003. This falsifies the simple H-OCR explanation for CLIP's zero-shot edge: if CLIP were winning by reading page titles, removing the title region would tank its retrieval, but the opposite happens. The rise is consistent with the title region acting as a low-information template (every Wikipedia page begins with the same chrome), and removing it yields cleaner content-driven matches.

Table 5. Phase-2 cross-set R@1 with the top fraction of every anchor and pool image painted white. The same fine-tuned checkpoints that lose Protocol-A R@10 under blanking (Table 4) *gain* Phase-2 R@1 by the same information removal. CLIP+MLP-30k embeddings collapsed; all predictions are identical.

Run	orig. R@1	b-15 R@1	b-30 R@1
DINOv2-MLP-30k	0.169	0.712	0.763
DINOv2-SALAD-30k	0.212	0.686	0.737
DINOv2-SALAD-50k	0.110	0.661	0.737
CLIP-SALAD-30k	0.551	0.686	0.720
CLIP-MLP-30k†	0.932	0.932	0.932

† Collapsed model.

Table 6. Image-to-text retrieval R@k on a held-out 1 000-page set disjoint from the prior eval slice. Image embeddings are I-JEPA target features (1280-d); text embeddings are BERT[CLS] of the page title (768-d). The adapter is a single linear map $A \in R^{1280\times768}$ trained on a 4 000-page train split with InfoNCE (τ=0.07, AdamW η=10^{-3}, 3 k steps). Random R@1 = 10^{-3}.

Configuration	R@1	R@5	R@10	R@20
I-JEPA $\leftrightarrow$ BERT, no adapter	0.000	0.008	0.015	0.026
+ linear adapter A	0.038	0.123	0.197	0.287

Trained DINOv2 loses heavily from blanking. Every fine-tuned DINOv2 checkpoint drops between 0.16 and 0.29 R@10 when the title is removed. The 50 k saturation checkpoint, the strongest in-distribution model in the paper, falls from 0.915 to 0.673. The same 30 k MLP run that beats zero-shot CLIP zero-shot (0.787 vs. 0.779) collapses to 0.494 under blanking. CLIP+SALAD moves only −0.037, indicating CLIP's fine-tune did not learn the shortcut to anywhere near the same degree. Fine-tuning on a single visual corpus has taught DINOv2 to bind page identity to the title-bearing top region.

Phase-2 reverses the sign on the same checkpoints. The title-region features that boost in-distribution Protocol-A are the very confound that hurts cross-set Phase-2 transfer (Table 5). Painting the top 30 % white yields up to 6.7× Phase-2 R@1 improvement on the same DINOv2-SALAD-50k that loses 0.242 R@10 in-distribution. Together, the two tables locate the failure precisely: it is not that the encoder cannot read the body region, it is that the title region is so reliably informative on Wikipedia that the model bypasses the body. Random title-region masking during training is the obvious mitigation.

4.5. I-JEPA + linear text adapter

The zero-shot result (Table 1) leaves Q2 at a sharp verdict: I-JEPA does not produce text-retrievable image representations out of the box. We test a softer version of Q2: do I-JEPA features encode any latent text-aligned structure that a small post-hoc adapter could recover? Following the recipe of (**?**) but on document screenshots, we train a single linear projection A from the I-JEPA target encoder (1280-d) into the BERT[CLS] text space (768-d) by InfoNCE on 4 000 (page-screenshot, page-title) pairs and evaluate image-to-title retrieval on a disjoint 1 000-page set (Table 6).

The bare cross-encoder retrieval is at chance (≤ 0.026 at R@20). The single linear adapter lifts R@10 from 0.015 to 0.197 — a 13$\times$ improvement from a fixed 1280$\times$768 projection that costs no nonlinearity and no extra parameters in the encoder. The conclusion qualifies Q2: I-JEPA features are not zero-shot text-aligned, but they encode features that BERT can recognise after a low-capacity linear correction. The Platonic Hypothesis prediction — that diverse pretraining objectives converge on a shared abstract geometry — holds in a weak form for I-JEPA on this domain, but the projection from one geometry to the other is non-trivial and must be learned.

5. Discussion and Limitations

Single seed. All training runs use one random seed. The qualitative ordering across encoder families is large enough to read past seed noise (the closest comparison, DINOv2-MLP-30k vs. DINOv2-SALAD-30k at ΔR@10 $= 0.093$, is well outside typical single-seed jitter), but the within-family numbers (CLIP-SALAD vs. CLIP-MLP at the same LR) require multi-seed reruns before strong claims.

Pool size. The headline retrieval pool ($P = 2000$) is small relative to industrial document corpora (10^5–10^6). The Phase-2 pool ($\sim$7 candidates per anchor) is smaller still. Our numbers should not be directly compared with ColPali-on-ViDoRe (**?**); the meaningful comparisons are within-table between encoders evaluated on identical protocols.

Crop resolution. At 224×224 px input, body text is subpixel. Anything called "reading" in this paper refers to heading-level text and layout structure, not fluent body content. A higher-resolution variant would change which signals are available and is a natural follow-up.

Adapter ceiling. The I-JEPA linear adapter (Table 6) reaches R@10 $= 0.197$ on the held-out set. This is well below CLIP zero-shot (0.779 on the larger Protocol-A pool with $P = 2000$), so "rescuing I-JEPA to CLIP-level retrieval" would require either a non-linear adapter, larger paired-data training, or both. The 0.197 figure is best read as evidence of *some* latent alignment, not as a practical recipe.

6. Conclusion

We compared six image encoders trained with five distinct pretraining objectives on a leak-free crop$\rightarrow$page retrieval task over Wikipedia screenshots. Image–text contrastive pretraining (CLIP, SigLIP) dominates zero-shot retrieval over image-only self-supervision (DINOv2, I-JEPA), supervised classification (ImageNet ViT), and random initialisation. Across the six encoders, alignment with text encoders rank-correlates with retrieval performance at Spearman $\rho = 0.83$, supporting the Platonic Representation Hypothesis on this domain. Fine-tuning DINOv2 with the SALAD aggregator on 50 k pages reaches R@10 $= 0.915$, exceeding zero-shot CLIP, but introduces a universal pathology: the fine-tuned model overweights the title region, dropping in-distribution Protocol-A R@10 by up to 0.29 under blanking while *improving* cross-set Phase-2 R@1 by up to 6.7$\times$. Zero-shot CLIP and SigLIP show the opposite sign under the same blanking, falsifying the simple title-OCR explanation for their zero-shot edge. A 1280$\times$768 linear adapter trained on (image, BERT-title) pairs lifts I-JEPA from near-chance to R@10 $= 0.197$, indicating image-only predictive features encode latent text-aligned structure that is not zero-shot accessible but is recoverable by a low-capacity linear projection.